Electrical & Computer ENGINEERING

Carnegie Mellon

Tentative Course Calendar

18-349/14-642: *Introduction to Embedded Systems*
Spring 2020

A	Class Activity	Lab Activity
1/14	L1: Introduction – Course Overview	
1/16	L2: PCB Lecture (Design)	Lab 0 (PCB) Out
1/21	L3: PCB Lecture 2 (Manufacturing)	
1/23	L4: ARM Architecture	Lab 0 Due, Lab 1 (Boot+ASM) Released
1/28	L5: ARM ASM Overview	
1/30	L6: ARM Profiling and Optimization	
2/4	L7: Memory Mapped I/O and Buses	
2/6	L8: Serial Protocols	Lab 1 Due, Lab 2 Released (Interfacing)
2/11	L9: Sampling, ADCs, DACs	
2/13	L10: Sensors and Actuators	
2/18	L11: Timers, Interrupts	
2/20	L12: SVC, Syscalls	Lab 2 Due, Lab 3 Release (Syscalls)
2/25	L13: Processes and Memory Management	
2/27	L14: Mid-term Review	
3/3	MIDTERM EXAM	
3/5	L15: Scheduling and Concurrency	
3/10	Spring Break - No Class	
3/12	Spring Break - No Class	
3/17	L16: Real-Time Scheduling 1-2	Lab 3 Checkpoint Due
3/19	L17: Real-Time Scheduling 2-2	
3/24	L18: RTOS Design and Implementation	Lab 3 Due, Lab 4 (RTOS Kernel) Released
3/26	L19: Multi-Core and SoC	
3/31	L20: Embedded Power Management	
4/2	L21: Real-Time Communication	Lab 4 Checkpoint Due
4/7	L22: Embedded Wireless Communication	
4/9	L23: Embedded Control	
4/14	L24: RT-Linux + LKM	L 4 Due, Lab 5 (System Integration) Released
4/16	Carnival – No Class	
4/21	L25: Safety Critical	
4/23	L26: Embedded Security	
4/28	L27: Advanced Topic	
4/30	L28: Course Wrap-up	Lab 5 Due
	FINAL EXAM PERIOD	